Dr. Sushovan Majhi

Assistant Professor of Data Science, George Washington University
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Research Interests

Applied Topology, **Topological Data Analysis (TDA)**, **Discrete and Computational Geometry**, **Shape Matching**, **Statistical Finance**, and **Topological Deep Learning**.

My research primarily revolves around the interface of *mathematics*, *computer science*, and the mathematical foundations of *data science*. More specifically, I am motivated to develop provable inference techniques for data science that are inspired by topology and geometry. I also keep a keen interest in applying TDA to fascinating, real-world problems arising in fields, like finance, dynamical systems, biology, medicine, and genetics.

Education

- **Doctor of Philosophy** in **Mathematics** — **August 2014–December 2020**
  Tulane University, New Orleans, LA, USA.
  Advisor: **Prof. Carola Wenk**
  Courses: computational geometry, computational topology, topological data analysis, differential geometry, differentiable manifolds, algorithms, data structures, computational complexity, applied mathematics, and scientific computing.

- **Master of Science** in **Mathematics** — **August 2009–May 2012**
  Tata Institute of Fundamental Research, Bangalore, India
  Courses: ordinary and partial differential equations, probability theory, complex analysis, functional analysis, numerical linear algebra, measure theory, mechanics.

- **Bachelor of Science** in **Mathematics** (Hons.) — **July 2006–May 2009**
  Ramakrishna Mission Vidyamandira, **Calcutta University**, West Bengal, India
  Courses: calculus, real analysis, linear algebra, numerical analysis, game theory, statistics, physics.

Work Experience

- **Assistant Professor** — **August 2024–*current***
  Data Science Program, George Washington University, Washington D.C., USA

- **Visiting Assistant Professor** — **August 2023–July 2024**
  Data Science Program, George Washington University, Washington D.C., USA

- **Postdoctoral Research Fellow** — **January 2021–July 2023**
  School of Information, University of California, Berkeley, USA
  Role: The responsibilities included conducting research broadly in data science, forging new research collaborations, organizing research webinar series.

Preprints

9. **SM**. A constant-factor approximation of the Gromov–Hausdorff distance in the plane, June 2026. Available at: arXiv:2606.17051 [cs.CG]

8. **SM**, Atish Mitra, Žiga Virk, Pramita Bagchi. A Closed-Form Adaptive-Landmark Kernel for Certified Point-Cloud and Graph Classification, May 2026. Available at: arXiv:2605.04046 [cs.LG]
   • Submitted to *Journal of Machine Learning Research*

7. **SM**, Atish Mitra, Žiga Virk, Pramita Bagchi. A Closed-Form Persistence-Landmark Pipeline for Certified Point-Cloud and Graph Classification, May 2026. Available at: arXiv:2605.02836 [cs.LG]
   • Submitted to *Transactions on Machine Learning Research*

6. Salam Rabindrajit Luwang, **SM**, Vishal Mandal, Atish J. Mitra, Md. Nurujjaman, Buddha Nath Sharma. Interpretable Classification of Time Series Using Euler Characteristic Surfaces , 25 pages, March 2026. Available at: arXiv:2603.15079 [cs.LG]

5. Kazuhiro Kawamura, **SM**, Atish Mitra. The Shadow of Vietoris–Rips Complexes in Limits, 29 pages, January 2026. Available at: arXiv:2601.01359 [math.AT]
   • Submitted to *Foundations of Computational Mathematics*

4. Joshua Dorrington, **SM**, Atish Mitra, James Moukheiber*, Demi Qin, Jacob Sriraman*, Kristian Strommen. Topology of the Polar Vortex and Montana Weather, 6 pages, Mar 2025. Available at: arXiv:2503.20743 [math.DS]
   • Presented at *ATMCS'25* & subitted to *La Matematica*

3. Henry Adams, **SM**, Fedya Manin, Nicolò Zava, Žiga Virk. Lower Bounding The Gromov–Hausdorff Distance In Metric Graphs, 20 pages, Nov 2024. Available at: arXiv:2411.09182v1 [math.MG]
   • Accepted to SoCG'26 & submitted to *Discrete and Computational Geometry*

2. Halley Fritze, **SM**, Marissa Masden, Atish Mitra, Michael Stickney. Faithful Reeb Graph Reconstruction of a Tectonic Subduction Zone from Earthquake Hypocenters. Available at: arXiv:2410.19410 [cs.CG]
   • Presented at *ATMCS'25* & submitted to *La Matematica*

1. Rafal Komendarczyk, **SM**, Will Tran. Topological Stability and Latschev-type Reconstruction Theorems for Spaces of Curvature Bounded Above, 27 pages, June 2024. Available at: arXiv:2406.04259 [math.AT]
   • Submitted to *Discrete & Computational Geometry*

Refereed Journal Publications

11. Rafal Komendarczyk, **SM**, Atish Mitra. Vietoris–Rips Shadow for Euclidean Graph Reconstruction. *Journal of Applied and Computational Topology*, vol. 10, no. 3, September 2026. DOI: 10.1007/s41468-026-00242-2. Also available at: arXiv:2506.01603 [math.AT]

10. Buddha Nath Sharma, Anish Rai, SR Luwang, Md. Nurujjaman, **SM**. Causality Analysis of COVID-19 Induced Crashes in Stock and Commodity Markets: A Topological Perspective. *International Journal of Modern Physics C*, September 2025. DOI: 10.1142/S0129183126500221. Also available at: arXiv:2502.14431 [q-fin.ST]

9. Henry Adams, Florian Frick, **SM**, Nichola McBride*. Hausdorff VS Gromov–Hausdorff Distances. *Discrete and Computational Geometry*, February 2025. DOI: 10.1007/s00454-025-00722-9 Also available at: arXiv:2309.16648[math.MG]

8. Kundan Mukhia, Anish Rai, SR Luwang, Md Nurujjaman, **SM**, Chittaranjan Hens. Complex Network Analysis of Cryptocurrency Market During Crashes. *Physica A: Statistical Mechanics and its Applications* , November 2024. DOI: 10.1016/j.physa.2024.130095. Also available at: arXiv:2405.05642 [q-fin.ST]

7. Anish Rai, Buddha Nath Sharma, Salam Rabindrajit Luwang, Md.Nurujjaman, **SM**. Identifying Extreme Events in the Stock Market: A Topological Data Analysis. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, October 2024. DOI: 10.1063/5.0220424. Also available at: arXiv:2405.16052 [q-fin.ST]

6. **SM**. Demystifying Latschev's theorem: Manifold Reconstruction from noisy data. *Discrete & Computational Geometry*, May 2024. DOI: 10.1007/s00454-024-00655-9. Also available at: arXiv:2305.17288[math.AT]

5. **SM** and Carola Wenk. Distance Measures for geometric graphs. *Computational Geometry: Theory and Applications*, March 2024. DOI: 10.1016/j.comgeo.2023.102056. Also available at: arXiv:2209.12869[cs.CG]

* Undergraduate student at the time of research or submission

4. **SM**. Vietoris–Rips complexes of metric spaces near a metric graph. *Journal of Applied and Computational Topology*, May 2023. DOI: 10.1007/s41468-023-00122-z. Available at: arXiv:2204.14234[math.AT]

3. **SM**, Jeffrey Vitter, and Carola Wenk. Approximating Gromov-Hausdorff distance in Euclidean space. *Computational Geometry: Theory and Applications*, January 2024. DOI: 10.1016/j.comgeo.2023.102034. Also available at: arXiv:1912.13008[math.MG]

2. Anish Rai, Ajit Mahata, Md Nurujjaman, **SM**, and Kanish Debnath. A sentiment-based modeling and analysis of stock price during the COVID-19: U- and Swoosh-shaped recovery. *Physica A: Statistical Mechanics and its Applications*. April 2022. DOI: 10.1016/j.physa.2021.126810. Also available at: arXiv:2110.03986 [q-fin.ST]

1. Brittany Terese Fasy, Rafal Komendarczyk, **SM**, and Carola Wenk. On the reconstruction of geodesic subspaces of $\mathbb{R}^N$. *International Journal of Computational Geometry & Applications*. September 2022. DOI: 10.1142/S0218195922500066. Also available at: arXiv:1810.10144[math.AT]

Refereed Conference Proceedings

4. Henry Adams, **SM**, Fedya Manin, Nicolò Zava, Žiga Virk. Lower Bounding The Gromov–Hausdorff Distance In Metric Graphs. In Proceedings of the *42nd International Symposium on Computational Geometry (SoCG)*, New Jersey, USA, June 2–5, 2026. Available at: arXiv:2411.09182v1 [math.MG]

3. **SM**. Demystifying Latschev's theorem: Manifold Reconstruction from noisy data. In Proceedings of the *40th International Symposium on Computational Geometry (SoCG)*, Athens, Greece, June 11–14, 2024. Available at: https://doi.org/10.4230/LIPIcs.SoCG.2024.73

2. Erin Chambers, Brittany Fasy, Benjamin Holmgren*, **SM**, and Carola Wenk. Metric and path-connectedness properties of the Frěchet distance for paths and graphs. In Proceedings of the *34th Canadian Conference on Computational Geometry (CCCG)*, 2023. DBLP: conf/cccg/ChambersHFMW23. Available at: arXiv:2308.00900[cs.CG]

1. **SM**. Graph mover's distance: An efficiently computable distance measure for geometric graphs. In Proceedings of the *34th Canadian Conference on Computational Geometry (CCCG)*, 2023. DBLP: conf/cccg/Majhi23. Available at: arXiv:2306.02133[cs.CG]

Workshop Contributions

8. Pramita Bagchi, **SM**, Atish Mitra, Žiga Virk. Statistical Analysis of Persistence Landmarks: A Hilbert Space Embedding of the Space of Persistence Diagrams, At *Fall Workshop on Computational Geometry (FWCG)*, Queens College, 2025.

7. Enrique Alvarado, Daniela Beckelhymer, Joshua Dorrington, Tung Lam, **SM**, Jasmine Noory, María Sánchez Muniz, Kristian Strommen. Detecting the Indian Monsoon using Topological Data Analysis. At *ATMCS'25*. June 2025. Available at: arXiv:2504.01022 [physics.ao-ph]

6. Halley Fritze, **SM**, Marissa Masden, Atish Mitra. Hausdorff–Approximation of Metric Graphs, At *Fall Workshop on Computational Geometry (FWCG)*, Tufts University, 2024. Available at: arXiv:2410.19410 [cs.CG]

5. Halley Fritze, **SM**, Marissa Masden, Atish Mitra. Hausdorff–Approximation of Metric Graphs, At *Fall Workshop on Computational Geometry (FWCG)*, Tufts University, 2024. Available at: arXiv:2410.19410 [cs.CG]

4. E. Chambers, B. Fasy, B. Holmgren* , **SM**, and C. Wenk. Path-Connectivity of Fréchet Spaces of Graphs. Computational Geometry: Young Researchers Forum, 2022

3. **SM** and Carola Wenk. Distance Measures for Geometric Graphs, At *Fall Workshop on Computational Geometry (FWCG)*, 2022

2. Brittany Terese Fasy, **SM** and Carola Wenk. Threshold-based graph reconstruction using discrete Morse theory. In *Fall Workshop on Computational Geometry (FWCG)*, 2018. Abstract available at: Link

1. Brittany Terese Fasy, Rafal Komendaczyk, **SM**, and Carola Wenk. Topological reconstruction of metric graphs in $\mathbb{R}^N$. At *Fall Workshop on Computational Geometry (FWCG)*, 2017. Abstract available at: Link

Graduate Teaching Experience

- **Linear Algebra for Data Science** **Summer 2024**
  Data Science Program, George Washington University, USA
  Topics: linear algebra foundations, Gaussian elimination, pivoting techniques for stability, LU decomposition, eigenvalues and eigenvectors, solving linear systems using iterative methods, projections and linear regression, SVD and applications. *Website:* `linalg.mathematics.land`

- **Algorithm Design (DATS 6001)** **Spring 2024–*current***
  Data Science Program, George Washington University, USA
  Topics: basics of computational complexity, data structures (array, stack, queue, tree, etc), algorithms (search, sort, etc), and programming paradigms like dynamic programming and greedy algorithms.
  Website: `6001-majhi.datasci.land`

- **Topological Data Analysis (CS)** **Fall 2023**
  National Institute of Technology, Sikkim, India
  Topics: topological spaces, metric spaces and their examples, simplicial complexes, homology in $\mathbb{Z}/2$ coeff, persistent homology, bottleneck and Wasserstein distance, and non-linear time-series analysis using TDA.
  Website: `tda.mathematics.land`

- **Computer Science Foundations (DATS 6450)** **Fall 2023**
  Data Science Program, George Washington University, USA
  Topics: computer design, programming in Python, object-oriented programming

- **Introduction to Data Mining (DATS 6103)** **Fall 2023–*current***
  Data Science Program, George Washington University, USA
  Topics: data wrangling, linear and logistic regression, classification, clustering, data visualization, support vector machines, machine learning algorithms.

- **Statistics for Data Science (MIDS 203)** **August 2020–July 2023**
  School of Information, University of California, Berkeley, USA
  Topics: probability theory, sampling distributions, estimators and convergence theorems, confidence intervals, hypothesis testing, and regression.

- **Linear Algebra, Complex Analysis** **January 2013–June 2013**
  Christ University and Scimetric Pvt Ltd, Bangalore, India

- **Analysis, Linear Algebra, Complex Analysis** **November 2011–July 2012**
  GATE-IIT Coaching Institute, JP Nagar, Bangalore, India
  Graduate level, for competitive national exams, e.g., National Eligibility Test

- **Analysis and Linear Algebra** **February 2012–July 2012**
  MES College, Department of Mathematics, Malleswaram, Bangalore, India

Undergraduate Teaching Experience

- **Data Science for All** **Summer 2025**
  Data Science Program, George Washington University, USA
  Topics: Foundations of data science for undergraduate students

- **Data Science Capstone (DATS 4001)** **Fall 2024–*current***
  Data Science Program, George Washington University, USA
  Goal: The goal of the capstone is for the students to apply the knowledge acquired throughout the GWU Data Science undergraduate program to solve a real-world problem, using real data. It typically starts with developing S.M.A.R.T. questions and goes through the entire data science life cycle in multiple iterations.

- **Undergraduate Statistics for Business Students (MATH 1140)** **Summer 2019**
  Tulane University, USA
  Topics: sampling methods, descriptive statistics, probability theory, random variables, limit theorems, confidence intervals, hypothesis testing, and linear regression.

Other Teaching

- **Graduate Teaching Assistant** **Fall 2014–Spring 2017**
  Tulane University, USA

Invited Talks and Accepted Abstracts

- **Title:** Lower Bounding the Gromov–Hausdorff Distance in Metric Graphs **June 5, 2026**
  42nd International Symposium on Computational Geometry (SoCG), Rutgers University, New Brunswick, NJ

- **Title:** Gemetric Latschev's Theorem **March 11, 2026**
  Spring Topology Conference, University of Alabama, D.C.

- **Title:** Vietoris–Rips Shadow for Euclidean Graph Reconstruction **January 7, 2026**
  Joint Mathematics Meetings, Washington, D.C.

- **Title:** Vietoris–Rips Shadow for Euclidean Graph Reconstruction **Nov 8, 2025**
  Fall Workshop on Computational Geometry, Queens College

- **Title:** Vietoris–Rips Shadow for Euclidean Graph Reconstruction **Oct 4, 2025**
  AMS Southeastern Sectional Meeting, Tulane University

- **Title:** Topological and Geometric Reconstruction of Spaces Beyond Manifolds **June 19, 2025**
  TDA Week, Kyoto University, Japan, 2025

- **Title:** Topological Stability and Latschev-type Reconstruction Theorems for CAT(κ) Spaces **March 8, 2025**
  AMS Southeastern Sectional Meeting, Clemson University

- **Title:** Topological Stability and Latschev-type Reconstruction Theorems for Spaces of Curvature Bounded Above **March 6, 2025**
  Spring Topology and Dynamics, Christopher Newport University, Newport News

- **Title:** Predicting the Onset and Withdrawal of the Indian Monsoon using Persistent Homology **January 9, 2025**
  AMS Special Session: Climate Science at the Interface Between Topological Data Analysis and Dynamical Systems Theory, Joint Mathematics Meetings, Seattle, 2025

- **Title:** Lower Bounding the Gromov–Hausdorff distance in Metric Graphs **Nov 14–15, 2024**
  The 31st Fall Workshop on Computational Geometry, Tufts University

- **Title:** A Taste of Topological Data Analysis (TDA): Reconstruction of Shapes **Oct 14, 2024**
  Indian Institute of Technology, Mandi, India

- **Title:** A Taste of Topological Data Analysis (TDA): Reconstruction of Shapes **Oct 8, 2024**
  Vellore Institute of Technology, Chennai, India

- **Title:** Demystifying Latschev's Theorem **June 14, 2024**
  Symposium on Computational Geometry, Athens, Greece
- **Title:** Demystifying Latschev's Theorem **March 28, 2024**
  Montana State University
- **Title:** Demystifying Latschev's Theorem **August 23, 2023**
  Applied Algebraic Topology Research Network (AATRN)
  Available on: YouTube
- **Title:** Graph Mover's Distance **August 2–4, 2023**
  The 34th Canadian Conference on Computational Geometry, Montreal, Canada
- **Title:** Similarity Measures for Geometric Graphs **October 14–15, 2022**
  The 30th Fall Workshop on Computational Geometry
  North Carolina State University, USA
- **Title:** Topological Methods in the Reconstruction and Comparison of Shapes **February, 2022**
  Mathematics Department, ICFAI University, Tripura, India
- **Title:** A Taste of Topological Data Analysis (TDA): Reconstruction of Shapes **September, 2021**
  Department of Mathematics, Hunter College, NY, USA
- **Fall Workshop on Computational Geometry** **October, 2018**
  Queens College, New York, USA
- **Fall Workshop on Computational Geometry** **November, 2017**
  SUNY (Stony Brook), New York, USA

PhD Dissertation Committee

- Peiqi Yang, Department of Mathematics, George Washington University **Summer 2025**
- Will Tran, Department of Mathematics, Tulane University, **Spring 2024**
  Thesis Title: Distortion and Curvature in the Shape Reconstruction Problem

PhD Advisees

1. Buddha Nath Sharma, PhD student at **August 2025–*current***
   National Institute of Technology, Sikkim, India
   Thesis Title: Applications of topological data analysis in financial markets
   Co-advised with Md. Nurujjaman

Graduate Student Mentees

- Mauricio Montes, current PhD student
  Auburn University **March 2025–August 2025**
  Project Title: Statistical Analysis of Euler Characteristic Surfaces
- Khush Shah and Shikha Kumari, was data science master's students
  George Washington University, **August 2023–December 2024**
  Project Title: Matching geometric graphs using Graph Neural Networks
  Now: Junior Data Scientist at Cintra and AI/ML & Software Engineer at Nexas 8 International, respectively
- Anish Rai, was a PhD student at **August 2022–August 2024**
  National Institute of Technology, Sikkim, India
  Project Title: Prediction of stock market crashes using topological data analysis
  Now: Postdoctoral scholar at Chennai Mathematical Institute (CMI), Chennai, Bangalore

Undergraduate Student Mentees

- Abby Stein
  Senior at George Washington University **August 2025–*current***

- Jacob Sriraman
  was at Montana State University **August 2025–*current***
  Now: Master's student at Montana State University

- James Moukheiber
  was at George Washington University **August 2024–*current***
  Now: Master's student at the University of Zurich

Department/University Service

- Chair, Undergraduate Curriculum Committee
  George Washington University, Data Science Program **August 2025–*current***

- Undergraduate Advisor
  George Washington University, Data Science Program **August 2024–*current***

Awards, Scholarships, Recognition

- **Travel Grant**, University of California, Berkeley
- **UGC-CSIR NET Research Fellowship**, India, June 2012
- **TIFR Junior Research Fellowship** for pursuing Integrated PhD studies at TIFR-CAM, Bangalore, India. August 2009.
- **Secured grade "A" in SCIENCE TALENT SEARCH EXAMINATION**
  conducted by JATIYA VIJNAN PARISAD and INDIAN SCIENCE CONGRESS ASSOCIATION

Academic Services

- I organized two special sessions on "Topological Data Analysis for Non-linear Dynamics" and "Topological and Geometric Shape Reconstruction" at the Joint Mathematics Meetings (JMM) in Washington DC, 2025.
- I co-organized a special session on "Advances in Applied Topology and Topological Data Analysis" at the AMS southeastern sectional meeting at Tulane University in 2025.
- I have been a reviewer for journals, like Discrete and Computational Geometry, Foundations of Data Science, Journal of Combinatorial Optimization, and International Journal of Computational Geometry.
- I have been a reviewer for conferences, like the International Symposium on Computational Geometry, International Symposium on Spatial and Temporal Databases, European Workshop on Computational Geometry, ACM International Conference on Advances in Geographic Information Systems, WADS Algorithms and Data Structures Symposium.
- I organized SIAM Graduate Student Chapters at Tulane University.
- I have been organizing a data science webinar series at the University of California, Berkeley.

Computational Skills

Java, C, R, Python, Ruby, JavaScript, SQL, Bash.

Software Projects

- **Simplicial Complexes in JS** GitHub
  JavaScript implementation of some of the widely used computations on simplicial complexes. The library also implements the Smith Normal Form in order to compute the homology groups of an abstract complex.

- **Shape Reconstruction Visualization** WebApp | GitHub
  To complement my PhD research, I implemented my topological reconstruction algorithm for planar metric graphs in this library. The library is written in JavaScript and made available to users as a web-app.
  Skills: JavaScript, HTML, CSS.

Popular Media

- Interviewed by GWU Hatchet for an Article **August 2024**

Entrepreneurial Experience

- **Scimetric Edulabs Private Limited** **December 2012–April 2017**
  Bangalore, India
  Role: **co-founder** and **director**
  In this start-up venture, our objective was to motivate and train students in higher education. We won the franchise to work with several private colleges in India. We coached science students for standardized entrance tests for PhD and academic jobs. The company employed 6 trainers.

Hobby

In my spare time, I write tutorials on *random* topics in order to make mathematics and statistics a little more interactive; they can be found here: https://www.smajhi.com/tutorials. I also enjoy playing the piano and classical guitar.